

Sowmya Wilson

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SUMMARY

Senior Software Engineer with 8 years building and owning high-throughput, event-driven backend systems in Java, Spring Boot, Kafka, and AWS. Owns a business-critical collections pipeline in a regulated financial environment, with a track record of large-scale data migrations, zero-downtime cutovers, and measurable gains in reliability and compute efficiency.

SKILLS

Languages	Java, Python, JavaScript, Lua, Bash, SQL
Frameworks	Spring Boot, Spring Batch, Spring Cloud, Spring Security, Spring AOP, Hibernate
Cloud & Data	AWS (S3, Lambda, EMR, EC2, ECS, Secrets Manager), Snowflake, Databricks
Streaming & Messaging	Kafka, RabbitMQ
DevOps	Docker, Kubernetes, Jenkins, Maven, Git, GitLab, CI/CD
Observability	New Relic, Splunk, Datadog, Grafana
Testing	JUnit, Mockito, Cucumber, Postman
Databases	MySQL, PostgreSQL, MongoDB

EXPERIENCE

Capital One — Senior Software Engineer

July 2024 – Present

- Own and maintain a business-critical, event-driven batch pipeline (S3 drop → Lambda → EMR → Spring Batch) that generates collections offers for customers in financial hardship; a single failed run triggers a Sev5, blocking offers and leaving agents with inaccurate customer data.
- Delivered feature development across the pipeline's EMR and Spring Batch stages, processing 16M+ records per run while preserving the data accuracy that downstream agent tooling depends on.
- Architected and delivered a high-throughput Kafka ingestion platform integrating Discover and Capital One collections systems, enabling real-time eligibility decisions and contributing to a reduction in delinquent accounts.
- Built a distributed Snowflake-to-Snowflake migration framework on Databricks, processing 90M+ records with high concurrency; increased compute efficiency by 125% and eliminated memory bottlenecks while ensuring post-merger data integrity.
- Operated as technical lead during a high-stakes merger integration, owning onboarding of Discover's email platform into Capital One with uninterrupted customer communications.
- Led a zero-downtime migration of secrets management from HashiCorp Vault to AWS Secrets Manager, improving security posture and accelerating credential rotation.
- Owned incident response across a portfolio of 9 backend applications triage, root cause analysis, hotfix deployment reducing MTTR; built proactive New Relic and Splunk alerting that shifted detection from reactive to preemptive.
- Mentored engineers and set code quality standards through design reviews and best practices in Java, Spring Boot, and distributed systems.

Toshiba GCS — Software Developer II

July 2021 – March 2024

- Designed and built Point-of-Sale (POS) applications in Spring Boot, delivering 10+ production features spanning order management, manager overrides, and financial transactions (gift-card cash-out, cash reconciliation).
- Engineered an automated license-verification feature to enforce regulatory compliance during sales transactions.
- Architected a cash-management protocol reconciling cash flow between the main safe and individual registers, improving security and accuracy.
- Built and maintained microservices and CI/CD pipelines (Java, Spring Boot, Jenkins), with unit testing via JUnit/Mockito and message processing through RabbitMQ.
- Raised logging and debugging standards using Splunk, and mentored engineers and interns through knowledge-transfer sessions.

Capital One — Software Developer

December 2018 – July 2021

- Developed and supported Spring Boot microservices and RESTful APIs on AWS (EC2, ECS, S3), owning build, deployment, and production support.
- Optimized API performance with custom Lua/Nginx modules and LRU caching (lua-resty-lrucache), reducing response times and improving throughput.
- Built internal developer tooling in Go (tdm-swipe, Bogie Infra Builder) and designed a GitHub pull-request validation pipeline in Jenkins to catch regressions before merge.
- Established observability with Datadog, Grafana, and Splunk, and defined infrastructure for QA, Performance, and Production environments.

EARLIER EXPERIENCE

- Java Developer, Cogent InfoTech** (Sep 2018 – Nov 2018) — Built RESTful web services and microservices with Spring and Hibernate.
- Software Developer Intern, Cell-Gate** (Jun 2018 – Aug 2018) — Built web-based tooling for a security camera and video-surveillance product using Java EE, Spring Boot, and Hibernate.
- Research Assistant, University of Southern Mississippi** (Sep 2016 – May 2018) — Applied ML (K-Means, classification) to NASA LIDAR/RADAR data in Python; co-authored research.

EDUCATION

M.S., Computer Science — University of Southern Mississippi, USA

May 2018

B.E., Electrical Engineering — Anurag Group of Institutions (JNTU), India

May 2016